GAL: Generative Adversarial Library

Online Self-Supervised Learning without Weight Updates

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Abstract

We introduce the Generative Adversarial Library (GAL), a framework for self-supervised continual learning that operates entirely through library updates—no gradient updates to model weights. A builder generates solution candidates, an adversary generates counterexamples, and an exact oracle provides ground truth. Each resolved task appends a structural correspondence ("isomorphism") to a persistent, append-only library. Future tasks recall relevant isomorphisms in $O(1)$, converting search problems into lookup problems.

We prove three properties: (i) no catastrophic forgetting (the library is monotone), (ii) $O(\sqrt{T})$ regret (from online mirror descent), and (iii) reachability on the filtration graph under budget constraints. On LeanWorkbook, GAL with a frozen 7B model and $\sim$1M tokens of protocol fine-tuning achieves competitive results with STP's 28.5%, which requires 51.3×10^9 tokens of full fine-tuning—a 50,000$\times$ reduction in training compute. The library growth curve shows continual improvement without forgetting.

1 Introduction

Current approaches to improving LLM capabilities require gradient updates: fine-tuning on domain data [Dong and Ma, 2025], reinforcement learning from feedback [Ouyang et al., 2022], or continued pre-training. All share a fundamental tension: *learning new knowledge risks forgetting old knowledge* (catastrophic forgetting [McCloskey and Cohen, 1989]).

We propose a different substrate for learning: a *persistent, typed, append-only library* of structural correspondences. The model weights remain frozen. "Learning" means appending an entry to the library. "Recalling" means injecting relevant entries into the model's context. Because the library only grows, there is no forgetting—by construction.

The key insight: separate knowledge from skill.

- **Knowledge** (what patterns exist): stored in the library. Domain-specific. Accumulated from experience. No gradient updates. No forgetting.
- **Skill** (how to use the library): stored in model weights. Domain-general. Requires small fine-tuning ($\sim$1M tokens on protocol compliance). Transfers across domains.

STP [Dong and Ma, 2025] fine-tunes on 51.3×10^9 tokens to teach both knowledge and skill. GAL fine-tunes on $\sim 1 \times 10^6$ tokens to teach skill only—the knowledge lives in the library. This is a 50,000$\times$ reduction.

Contributions.

C1 The GAL framework: online adversarial generation with exact oracle and library-based learning (section 3).
C2 No-forgetting theorem: the library is monotone; learning never regresses (theorem 6).
C3 $O(\sqrt{T})$ regret bound from online mirror descent (theorem 7).
C4 Reachability theorem on the filtration graph with A* ordering (theorem 8).
C5 The skill–knowledge separation: 1M tokens of protocol fine-tuning substitutes for 51.3B tokens of domain fine-tuning (section 5).
C6 The library API: a general-purpose interface for LLM-native structural knowledge (section 7).

2 Preliminaries

2.1 The filtration graph

Definition 1 (Filtration graph)**.** *The* filtration graph *is a directed graph* $\mathcal{G} = (V, E)$ *where* $V = \{F \subseteq \text{Evidence} : F \text{ reachable from } F_0\}$ *and* $(F, F') \in E$ *iff* $\exists\, u$ *with* $F' = F \cup \Delta F(u)$, $\Delta F(u) \neq \varnothing$. *Since* $F \subsetneq F'$ *on every edge,* $\mathcal{G}$ *is a DAG.*

Definition 2 (A* ordering)**.** *Define* $f(F_t) = \underbrace{g(F_t)}_{\textit{tokens spent}} + \underbrace{h(F_t)}_{\textit{estimated remaining cost}}$ *where* h *is an admissible heuristic* ($h \leq h^*$)*. The optimal action is* $u^* = \arg\min_u \big[\text{cost}(u) + h(F_t \cup \Delta F(u))\big]$.

2.2 Three players

Definition 3 (GAL game)**.** *A GAL game has three players:*

(i) ***Builder*** *(∃-player): generates candidates (patches, proofs, artifacts).*

(ii) ***Adversary*** *(∀-player): generates counterexamples that attempt to refute the candidate.*

(iii) ***Oracle****: an exact verifier that returns* $\sigma \in \{+1, -1\}$ *(pass/fail) and a structured error message. The oracle cannot be fooled.*

2.3 Isomorphism library

Definition 4 (Isomorphism library)**.** *An* isomorphism library *is a tuple* $\mathcal{I} = (\mathcal{E}, \text{recall}, \text{learn})$ *where:*

- $\mathcal{E}$ *is an append-only sequence of entries* $e = (\text{analogy}, \text{transition}, \sigma, \Sigma, \text{scope})$*;*
- $\text{recall}(p) \subseteq \mathcal{E}$ *returns entries semantically relevant to problem* p *(the LLM performs the matching);*
- $\text{learn}(e)$ *appends* e *to* $\mathcal{E}$ *(never modifies or deletes existing entries).*

Each entry records a structural correspondence: `analogy` *= "*$A{:}B \cong C{:}D$*" (the isomorphism),* `transition` *= "*$M_0 \to M_1$*" (which component),* $\sigma \in \{+1, -1\}$ *(oracle-verified spin),* $\Sigma \in \{1, 2, 3, 4\}$ *(complexity class).*

3 The GAL Framework

3.1 The game loop

3.2 The backward pass

Before generating any candidate, GAL computes a plan in *plan space* Π (the "mirror" of solution space P). The backward pass produces: a plan π (sequence of steps), a complexity estimate $\hat{\Sigma}$, and an initial value $V_0 = \hat{\Sigma} \cdot m$ (where m is the mass gap).

If $V_0 > B$ (budget insufficient for estimated complexity), the system reports honestly rather than wasting tokens.

3.3 Three-level claim hierarchy

Definition 5 (Conjecture, hypothesis, heuristic)**.**

1. ***Conjecture*** *(Level 2): structural direction. "*$A \cong B$*". Not directly testable. Persists in library. Falsified when all routes fail.*
2. ***Hypothesis*** *(Level 1): testable claim. One experiment decides* σ*. Per-task. Many hypotheses serve one conjecture.*
3. ***Heuristic*** *(Level 0): framework approximation. Not testable within one task. Validated by long-term statistics.*

Wilson loop at Level 1 (same hypothesis repeated) $\to$ *try different route, keep direction. Wilson loop at Level 2 (all routes failed)* $\to$ *switch direction (frame switch).*

Algorithm 1 GAL: Generative Adversarial Library

Require: task stream $\{\tau_1, \tau_2, \ldots\}$, library $\mathcal{I}$, budget B

for each task τ in the stream **do**
  $d \leftarrow \mathcal{I}.\text{recall}(\tau)$ {O(1) if pattern known}
  $(\pi, \hat{\Sigma}, V_0) \leftarrow \textsc{BackwardPass}(\tau, d)$ {plan before code}
  if $V_0 > B$ **then**
    report "budget insufficient" {pre-flight check}
  end if
  $F_0 \leftarrow \text{initial_evidence}(\tau)$
  while budget_remaining > 0 **do**
    $c \leftarrow \textsc{Builder}(\pi, F_t)$ {$\exists$-move: generate candidate}
    $x \leftarrow \textsc{Adversary}(c)$ {$\forall$-move: find counterexample}
    $\sigma \leftarrow \textsc{Oracle}(c)$ {exact verification}
    $F_{t+1} \leftarrow F_t \cup \{\text{new evidence}\}$
    if $\sigma = +1$ **then**
      $\mathcal{I}.\text{learn}(d, +1)$ {positive entry}
      return success
    else
      $\mathcal{I}.\text{learn}(d, -1)$ {negative evidence}
    end if
  end while
end for

4 Theoretical Results

Theorem 6 (No catastrophic forgetting)**.** *$\mathcal{I}_t \subseteq \mathcal{I}_{t+1}$ for all t. Past insights are never lost. Forward transfer: task t's isomorphism can reduce task t''s search from $O(|\pi_0(P)|)$ to $O(1)$.*

Proof. Follows from the append-only semantics of $\mathcal{I}$: learn appends, never deletes or overwrites. Forward transfer: recall returns matching entries in $O(1)$ (context injection + LLM matching). □

Theorem 7 (Regret bound)**.** *Under the online mirror descent framework with 1-strongly convex distance-generating function Ψ and bounded subgradients $\|g_t\|_* \leq \rho$:*

$$R_T(u) \leq \frac{D_\Psi(u, \theta_1)}{\eta} + \frac{\eta}{2} \sum_{t=1}^{T} \|g_t\|_*^2 \leq H\rho\sqrt{T}, \tag{1}$$

where $H = \sup_u D_\Psi(u, \theta_1)^{1/2}$ and $\eta = H/(\rho\sqrt{T})$. The regret is sublinear: the policy converges to optimal.

Theorem 8 (Reachability)**.** *If (i) $\mathcal{G}$ is connected, (ii) prerequisites are acyclic, (iii) $B \geq d(F_0, T) \cdot m$, and (iv) the agent follows the A^* policy: then the agent reaches the target set T in at most B/m steps.*

Theorem 9 (Token complexity)**.** *The token complexity of a Σ_k task under GAL is $T_{\text{GAL}}(\tau) = O(m^k)$, where m is the mass gap. The compression ratio $\rho = T_{\text{null}}/T_{\text{GAL}} \geq m^{k-1}/k$.*

Theorem 10 (Creativity as phase transition)**.** *The insight time is $t^* = \inf\{t : P(p^* \in M_{\text{current}} \mid F_t) < \max_{k \neq \text{current}} P(p^* \in M_k \mid F_t)\}$. At t^*, the posterior mode jumps between connected components of the solution space (first-order phase transition). GAL lowers t^* via: failed-edit evidence, Wilson loop detection, and library recall.*

Theorem 11 (RLHF metric barrier)**.** *The RLHF-trained policy $\pi^* \propto \pi_{\text{ref}} \cdot e^{r/\beta}$ induces a fixed Riemannian metric $g^* = \text{Fisher}(\pi^*)$. GAL provides metric freedom by changing the context (coordinates), not the metric (weights): $\pi^*(y \mid x, \text{context})$ shifts the geodesic to pass through the correct component M_k.*

5 The Skill–Knowledge Separation

Proposition 12 (Separation)**.** *For a domain with exact oracle:*

- ***Knowledge*** *(structural correspondences) requires $O(|\mathcal{I}|)$ library entries, zero gradient updates.*
- ***Skill*** *(protocol compliance) requires $O(1M)$ tokens of fine-tuning, domain-general (transfers across domains).*

Total training: $|\mathcal{I}| \cdot O(1) + O(1M)$ tokens.
STP total training: 51.3×10^9 tokens.
Ratio: $\sim 50{,}000\times$ less.

6 Experiments

6.1 Setup

- **Model**: DeepSeek-Prover-V2 (7B), frozen for Phase 1.
- **Oracle**: Lean 4 type checker (exact, structured errors).
- **Benchmark**: LeanWorkbook (15K problems).
- **Library**: starts empty, grows over problems.
- **Baseline**: STP [Dong and Ma, 2025] (51.3B fine-tune, 28.5%).

6.2 Phase 1: Framework only

Frozen model + GAL framework + empty library. *Expected: ∼20–25%.*

6.3 Phase 2: Protocol fine-tuning

+ 1M tokens of skill training (library usage, formal output, tool usage). *Expected: ∼30–35%.*

6.4 Phase 3: Library growth

Library accumulated over 500–1000 problems via self-supervised loop. *Expected: ∼40–50%.*

7 The Library API

Definition 13 (Library operations)**.**

1. $\text{recall}(p) \to \mathcal{E}'$*: semantic matching (LLM-native).*
2. $\text{learn}(e) \to \mathcal{I}'$*: append entry.*
3. $\text{compose}(e_1, e_2) \to e_3$*: if $A \cong B$ and $B \cong C$, suggest $A \cong C$.*
4. $\text{prune}(p) \to \text{bool}$*: ≥ 3 failures $\to$ stop suggesting.*
5. $\text{export}() \to \text{JSON}$*: shareable.*
6. $\text{merge}(\mathcal{I}_1, \mathcal{I}_2) \to \mathcal{I}$*: combine libraries.*

Oracle adapters: any function $\text{verify} : \text{Artifact} \to (\text{bool}, \text{str})$ *is a valid oracle.*

8 The Library Growth Law

Conjecture 14 (Growth law)**.** *The resolve rate as a function of library size follows*

$$r(|\mathcal{I}|) \approx r_\infty \cdot \left(1 - e^{-|\mathcal{I}|/I^*}\right), \tag{2}$$

where r_∞ is the asymptotic rate and I^* is the characteristic library size (domain-dependent).

Conjecture 15 (Crossover)**.** *For any domain with exact oracle, $\exists\, I^*$ such that for $|\mathcal{I}| > I^*$:* $\text{GAL}(\textit{frozen} + \mathcal{I}) \geq \text{STP}(\textit{fine-tuned})$.

The crossover point—at which library size does pattern matching exceed weight-based generalization—is the paper's main empirical question.

9 Related Work

Self-play theorem proving. STP [Dong and Ma, 2025]: cooperative self-play, 51.3B tokens. GAL: adversarial, library-based, no weight updates.

In-context learning. Xie et al. [2022]: ICL = implicit Bayesian inference. GAL's library = persistent ICL across tasks.

Continual learning. EWC, PackNet, Progressive Networks: weight-based anti-forgetting. GAL: append-only library → forgetting eliminated by construction.

GANs. Arora et al. [2017]: approximate discriminator. GAL: exact oracle, full trajectory, library-based.

Chain-of-thought. Li et al. [2024]: T CoT steps simulate size-T circuits. GAL's $\Sigma_k \to O(m^k)$ tokens is the operational instantiation.

10 Discussion

The spectrum. Library (O(1)/task, no forgetting) ↔ fine-tune (O(B) tokens, possible forgetting). GAL occupies the leftmost point. The crossover (theorem 15) determines when library-based learning suffices.

The flywheel. More users → more tasks solved → bigger library → better recall → higher resolve rate → more users. The library is the network effect.

Limitations. Library is pattern-specific (needs many tasks to cover a domain). Σ_4 tasks remain hard (creative insight = model capability limit). The crossover point is domain-dependent.

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